AKSHAYA NARAYANASAMY

Los Angeles, CA | 614-769-8507 | akshaya25@g.ucla.edu
ORCID | Google Scholar | LinkedIn

PROFILE

Computational chemist and biologist developing integrative modeling and machine-learning methods for protein structure prediction and design. My research combines integrative modeling with experiments, deep learning, and Rosetta-based modeling to improve the structural modeling of proteins and protein complexes. My work also includes small molecule docking for Glioblastoma cancer targets, molecular dynamics simulations to characterize conformational energetics, and the design of antibacterial protein binders.

EDUCATION

University of California, Los Angeles **Expected 2027**
Ph.D. Student, Theory and Computational Chemistry | Department of Chemistry and Biochemistry | Los Angeles, CA

The Ohio State University **May 2025**
M.S. in Physical Chemistry | Department of Chemistry | Columbus, OH

Indian Institute of Science Education and Research, Thiruvananthapuram (IISER TVM) **July 2022**
B.S.-M.S. Major in Chemistry; Minor in Physics | Thiruvananthapuram, India

RESEARCH EXPERIENCE

Doctoral Researcher | Lindert Lab **Jan 2023–Present**
UCLA (July 2025–Present) | The Ohio State University (Jan 2023–June 2025)

- Developed integrative protein-complex modeling methods that combine AlphaFold/Rosetta predictions with ion-mobility mass spectrometry and cryo-EM data, leading two first author publications.
- Created IM-Complex, a Rosetta scoring approach that improved complex-model challenging targets relative to AlphaFold-Multimer.
- Developing experimental guided diffusion model.
- Applied molecular docking and virtual-screening workflows to identify candidate inhibitors for glioblastoma cancer-related protein targets.
- Built an AI based MurA binder-design pipeline, with potential antibacterial activity.

Visiting Student| Meiler Lab **May–June 2025**
Vanderbilt University | Nashville, TN

- Applied ProteinMPNN and ThermoMPNN to evaluate troponin C mutations and conformational energetics, leading to a first-author publication in the Journal of Chemical Information and Modeling.

Undergraduate Research Assistant | Hariharan Lab **Jan 2018–July 2022**
IISER TVM | Thiruvananthapuram, India

- Investigated how single-atom substitutions alter crystal packing, optical properties, and charge transport in organic chromophores using spectroscopy and computational analysis. The results lead to 3 publications including a co-first author paper.

Undergraduate Research Assistant | Shaji Lab **Apr–June 2021**
IISER TVM | Thiruvananthapuram, India

- Modeled quantum coherence and electron-transfer rates in a two-donor, one-acceptor system representing photosynthetic light harvesting.

PUBLICATIONS

6 total publications | 4 first/co-first-author publications|* Equal contribution.

1. **Narayanasamy, A.;** Drake, Z. C.; Turzo, S. M. B. A.; Rolland, A. D.; Prell, J. S.; Wysocki, V. H.; Lindert, S. Integrating Ion Mobility Mass Spectrometry Data with AlphaFold and Rosetta Improves Protein Complex Structure Prediction. *Analytical Chemistry* 2026. DOI: 10.1021/acs.analchem.6c01893
2. **Narayanasamy, A.;** Moretti, R.; Meiler, J.; Lindert, S. "Deep Learning Models Capture Umbrella Sampling-Derived Energetic Trends: A Troponin C Case Study." *Journal of Chemical Information and Modeling* 2026, 66(15), 9448–9457. DOI: 10.1021/acs.jcim.6c01925.

3. Howard, J. B.*; **Narayanasamy, A.*;** Lindert, S. "Improving Protein Structure Prediction Using Integrative Cryo-EM and Ion Mobility Mass Spectrometry Modeling." *Journal of Chemical Theory and Computation* 2026, 22(11), 5909–5917. DOI: 10.1021/acs.jctc.6c00265
4. John, A. T.; **Narayanasamy, A.;** Sudhakaran, K. P.; Hariharan, M. "Resonance-Assisted Hydrogen Bonding and π–π Stacking Modulates the Charge Transfer Coupling in Crystalline Naphthothiazoles." *Crystal Growth & Design* 2022, 22(9), 5686–5693. DOI: 10.1021/acs.cgd.2c00798
5. John, A. T.*; **Narayanasamy, A.*;** George, D.; Hariharan, M. "The Effect of Single-Atom Substitution on Structure and Band Gap in Organic Semiconductors." *Crystal Growth & Design* 2022, 22(2), 1237–1243. DOI: 10.1021/acs.cgd.1c01202
6. John, A. T.; **Narayanasamy, A.;** George, D.; Hariharan, M. "Atom-Efficient Halogen–Halogen Interactions Assist One-, Two-, and Three-Dimensional Molecular Zippers." *The Journal of Physical Chemistry C* 2021, 125(19), 10716–10722. DOI: 10.1021/acs.jpcc.1c01724

AWARDS, HONORS & RECOGNITION

Fellowship: **Audree V. Fowler Award in Protein Sciences,** MBI, UCLA **2026**
- *Selected as one of four recipients across multiple UCLA departments, awarded $10,000 to support research activities.*

Medal: **The Best Undergraduate Researcher Medal,** IISER Thiruvananthapuram **2022**
- *Awarded to one undergraduate student in the whole chemistry department who has good record of publications.*

Travel Awards: **NIH Biophysics Fellows Research Conference,** National Institutes of Health **2027**
Summer RosettaCon Travel Grant Award, RosettaCommons **2026**
Dr. Robin C. Burrell Memorial Fund Travel Award, The Ohio State University **2025**

Scholarship: **Central Sector Scheme of Scholarships for College and University Students,** India **2017–2022**

Cover Art: **Published Cover Art,** The Journal of Physical Chemistry C, 125(19) **2021**

Awards: **Chem Expo Prize Winner,** Anvesha, IISER Thiruvananthapuram **2017**
Class Topper Award, National Level Science Talent Search Examination, India **2012–2015**
- *Awarded to top students of each grade. Received every academic year.*

General Proficiency Award, St. Joseph of Cluny Matriculation Higher Secondary School, India **2007–2015**
- *Awarded to top students of each grade. Received every academic year.*

TEACHING & MENTORING

Course Development Teaching Assistant | CHEM 156: Physical Biochemistry **Mar–June 2026**
University of California, Los Angeles
- Developed iClicker questions, homeworks, and examinations; led weekly coding sessions.

Teaching Assistant | CHEM 1210: General Chemistry **August 2022–March 2023**
The Ohio State University
- Guided undergraduate students through recitations and laboratory experiments.

RosettaCommons RaMP Mentor **July 2024–July 2026**
UCLA (July 2025–Present) | The Ohio State University (July 2024–June 2025)
- Mentored postbaccalaureate researchers on Rosetta-based method development, data analysis, scientific communication, and graduate-school preparation.

Rotation Student Mentor **Aug 2023–Present**
The Ohio State University
- Trained rotation students in computational structural biology workflows and research practices.

PROFESSIONAL DEVELOPMENT

How to Train Your Model **August 1–8, 2026**
RosettaCommons | Seattle, WA
- Ion Mobility Guided Diffusion Model.

Rosetta Data Curation Hackathon **Mar 2–5, 2026**

RosettaCommons | San Juan, PR

- Curated machine-learning-ready protein datasets using AfCycDesign and PISCES.

C++ Bootcamp **Feb 1–8, 2024**

RosettaCommons | Vanderbilt University | Nashville, TN

- Completed an intensive program in C++ development for Rosetta applications.

Rosetta Machine Learning Bootcamp **Nov 17–21, 2023**

RosettaCommons | University of California, Davis | Davis, CA

- Developed a capstone workflow for optimization of a truncated β-glucosidase B enzyme.

SELECTED PRESENTATIONS

When AI is not enough: Guiding Computational Protein Structure Prediction with Experimental Data **10/2026**
Dr. Audree V. Fowler Fellowship Seminar, Molecular Biology Institute, University of California, Los Angeles, CA. *Talk*

Computation Structure Prediction Guided by Ion-Mobility Mass Spectrometry Data **08/2026**
Summer RosettaCon, Seattle, WA. *Talk*

Ion Mobility Mass Spectrometry Guided Modeling with AlphaFold and Rosetta Improves Protein Complex Structure Prediction **03/2026**
Seaborg Research Symposium, University of California, Los Angeles, CA. *Poster*

Ion Mobility Guided Monte Carlo-based Docking Pipeline for Predicting Protein Complexes **04/2025**
Ohio Supercomputer Center Research Symposium, OH *Talk*

Ion Mobility Guided Monte Carlo-based Docking Pipeline for Predicting Protein Complexes **03/2025**
Spring American Chemical Society Meeting, San Diego, CA. *Talk*

Ion Mobility Guided Monte Carlo-based Docking Pipeline for Predicting Protein Complexes **02/2025**
Winter RosettaCon, Boston, MA. *Talk*

TECHNICAL SKILLS

Programming & HPC	Python; shell scripting; Linux; Git; SSH; Slurm; PBS; SGE
Machine Learning	PyTorch; scikit-learn; NumPy; pandas; SciPy; Matplotlib; Seaborn; CNNs; GNNs; clustering
Structure & Design	Rosetta; AlphaFold; Chai; Boltz; ESMFold; RFdiffusion; ProteinMPNN; Protenix
Docking & Simulation	Glide; AutoDock; DOCK; GROMACS; NAMD; MDAnalysis
Visualization	Blender, PyMOL; VMD; UCSF Chimera